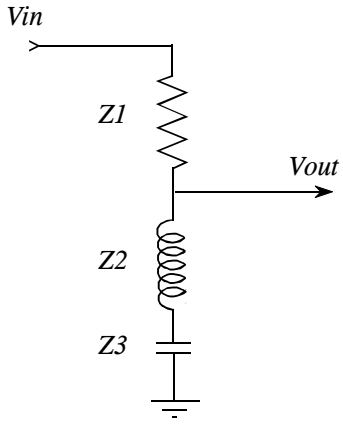

FIGURE 30-6
RLC notch filter. This circuit removes a narrow band of frequencies from a signal. The use of complex substitution greatly simplifies the analysis of this and similar circuits.

and inductors is treated the same as *resistance* in a DC circuit. This includes all of the basic combinations, such as: resistors in series, resistors in parallel, voltage dividers, etc.

As an example, Fig. 30-6 shows an *RLC* circuit called a **notch filter**, used to remove a narrow band of frequencies. For instance, it could eliminate 60 hertz interference in an audio or instrumentation signal. If this circuit were composed of three resistors (instead of the resistor, capacitor and inductor), the relationship between the input and output signals would be given by the voltage divider formula: $v_{out}/v_{in} = (R2+R3)/(R1+R2+R3)$. Since the circuit contains capacitors and inductors, the equation is rewritten with impedances:

$$\frac{Vout}{Vin} = \frac{Z2 + Z3}{Z1 + Z2 + Z3}$$

where: *Vout*, *Vin*, *Z1*, *Z2*, and *Z3* are all complex variables. Plugging in the impedance of each component:

$$\frac{Vout}{Vin} = \frac{j\omega L - \dfrac{j}{\omega C}}{R + j\omega L - \dfrac{j}{\omega C}}$$

Next, we crank through the algebra to separate everything containing a *j*, from everything that does not contain a *j*. In other words, we separate the equation into its real and imaginary parts. This algebra can be tiresome and long, but the alternative is to write and solve *differential equations*, an